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*      STAAD.Pro
*      Version 2007 Build 04
*      Proprietary Program of
*      Research Engineers, Intl.
*      Date= DEC 18, 2012
*      Time= 16:29:53
*
*      USER ID: Parsons Brinckerhoff
*****
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1. STAAD SPACE
INPUT FILE: 121211 FLOOR JOIST.STD
2. START JOB INFORMATION
3. ENGINEER DATE 29-AUG-12
4. END JOB INFORMATION
5. INPUT WIDTH 79
6. UNIT METER KN
7. JOINT COORDINATES
8. 1 0 0 0; 2 1.6 0 0
9. MEMBER INCIDENCES
10. 1 1 2
11. DEFINE MATERIAL START
12. ISOTROPIC STEEL
13. E 2.05E+008
14. POISSON 0.3
15. DENSITY 76.8195
16. ALPHA 1.2E-005
17. DAMP 0.03
18. END DEFINE MATERIAL
19. MEMBER PROPERTY JAPANESE
20. 1 TABLE ST C150X75X6.5
21. CONSTANTS
22. MATERIAL STEEL ALL
23. SUPPORTS
24. 1 PINNED
25. 2 FIXED BUT MY MZ KMX 0.00099
26. LOAD 1 LOADTYPE NONE TITLE DEAD LOAD: DL
27. MEMBER LOAD
28. 1 UNI GY -0.75
29. LOAD 2 LOADTYPE NONE TITLE LIVE LOAD: LL
30. MEMBER LOAD
31. 1 UNI GY -5
32. LOAD COMB 3 1.0DL + 1.0LL
33. 1 1.0 2 1.0
34. LOAD COMB 4 1.4DL + 1.6LL
35. 1 1.4 2 1.6
36. PERFORM ANALYSIS
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P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS = 2/ 1/ 2

SOLVER USED IS THE IN-CORE ADVANCED SOLVER

TOTAL PRIMARY LOAD CASES = 2, TOTAL DEGREES OF FREEDOM = 6

37. PRINT SECTION DISPL ALL

MEMBER SECTION DISPLACEMENTS

UNIT =INCHES FOR FPS AND CM FOR METRICS/SI SYSTEM

MEMB LOAD GLOBAL X,Y,Z DISPL FROM START TO END JOINTS AT 1/12TH PTS

1	1	0.0000	0.0000	0.0000	0.0000	-0.0009	0.0000
		0.0000	-0.0018	0.0000	0.0000	-0.0026	0.0000
		0.0000	-0.0031	0.0000	0.0000	-0.0035	0.0000
		0.0000	-0.0036	0.0000	0.0000	-0.0035	0.0000
		0.0000	-0.0031	0.0000	0.0000	-0.0026	0.0000
		0.0000	-0.0018	0.0000	0.0000	-0.0009	0.0000
		0.0000	0.0000	0.0000			
2		0.0000	0.0000	0.0000	0.0000	-0.0063	0.0000
		0.0000	-0.0122	0.0000	0.0000	-0.0171	0.0000
		0.0000	-0.0209	0.0000	0.0000	-0.0232	0.0000
		0.0000	-0.0240	0.0000	0.0000	-0.0232	0.0000
		0.0000	-0.0209	0.0000	0.0000	-0.0171	0.0000
		0.0000	-0.0122	0.0000	0.0000	-0.0063	0.0000
		0.0000	0.0000	0.0000			
3		0.0000	0.0000	0.0000	0.0000	-0.0073	0.0000
		0.0000	-0.0140	0.0000	0.0000	-0.0197	0.0000
		0.0000	-0.0240	0.0000	0.0000	-0.0267	0.0000
		0.0000	-0.0276	0.0000	0.0000	-0.0267	0.0000
		0.0000	-0.0240	0.0000	0.0000	-0.0197	0.0000
		0.0000	-0.0140	0.0000	0.0000	-0.0073	0.0000
		0.0000	0.0000	0.0000			
4		0.0000	0.0000	0.0000	0.0000	-0.0114	0.0000
		0.0000	-0.0220	0.0000	0.0000	-0.0310	0.0000
		0.0000	-0.0378	0.0000	0.0000	-0.0421	0.0000
		0.0000	-0.0435	0.0000	0.0000	-0.0421	0.0000
		0.0000	-0.0378	0.0000	0.0000	-0.0310	0.0000
		0.0000	-0.0220	0.0000	0.0000	-0.0114	0.0000
		0.0000	0.0000	0.0000			

MAX LOCAL DISP = 0.04351 AT 80.00 LOAD 4 L/DISP= 3677

***** END OF SECT DISPL RESULTS *****

38. PRINT MEMBER FORCES

MEMBER END FORCES STRUCTURE TYPE = SPACE

ALL UNITS ARE -- KN METE (LOCAL)

MEMBER	LOAD	JT	AXIAL	SHEAR-Y	SHEAR-Z	TORSION	MOM-Y	MOM-Z
1	1	1	0.00	0.60	0.00	0.00	0.00	0.00
		2	0.00	0.60	0.00	0.00	0.00	0.00
	2	1	0.00	4.00	0.00	0.00	0.00	0.00
		2	0.00	4.00	0.00	0.00	0.00	0.00
	3	1	0.00	4.60	0.00	0.00	0.00	0.00
		2	0.00	4.60	0.00	0.00	0.00	0.00
	4	1	0.00	7.24	0.00	0.00	0.00	0.00
		2	0.00	7.24	0.00	0.00	0.00	0.00

***** END OF LATEST ANALYSIS RESULT *****

39. PARAMETER 1
40. CODE BS5950
41. TRACK 2 ALL
42. BEAM 2 ALL
43. CHECK CODE ALL

STAAD.Pro CODE CHECKING - (BSI)

PROGRAM CODE REVISION V2.12_5950-1_2000

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
1 ST	C150X75X6.5	PASS	BS-4.3.6	0.096	4
		0.00	0.00	2.90	0.00

MATERIAL DATA

Grade of steel = S 275
 Modulus of elasticity = 205 kN/mm2
 Design Strength (py) = 275 N/mm2

SECTION PROPERTIES (units - cm)

Member Length = 160.00
 Gross Area = 23.71 Net Area = 23.71 Eff. Area = 23.71

	z-z axis	y-y axis
Moment of inertia	: 861.000	117.000
Plastic modulus	: 134.000	45.600
Elastic modulus	: 114.800	22.414
Effective modulus	: 134.000	45.600
Shear Area	: 13.500	9.750

DESIGN DATA (units - kN,m)

Section Class : PLASTIC

	z-z axis	y-y axis
Moment Capacity	: 36.9	9.2
Reduced Moment Capacity	: 36.9	9.2
Shear Capacity	: 222.8	160.9

BUCKLING CALCULATIONS (units - kN,m)

(axis nomenclature as per design code)

LTB Moment Capacity (kNm) and LTB Length (m): 30.31, 1.600
 LTB Coefficients _Associated Moments (kNm):
 mLt = 1.00 : mx = 0.00 : my = 0.00 : myx = 0.00
 Mlt = 2.90 : Mx = 0.00 : My = 0.00 : My = 0.00

CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

CLAUSE	RATIO	LOAD	FX	VY	VZ	MZ	MY
BS-4.2.3-(Y)	0.045	4	-	7.2	-	-	-
BS-4.3.6	0.096	4	-	7.2	-	2.9	-

Torsion and deflections have not been considered in the design.

***** END OF TABULATED RESULT OF DESIGN *****

44. FINISH

***** END OF THE STAAD.Pro RUN *****

**** DATE= DEC 18,2012 TIME= 16:29:53 ****

* For questions on STAAD.Pro, please contact *

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